Supplementary data 1

PROSPERO Protocol

Citation

Vicky Joshi, Jan Christensen, Esben Lejsgaard, Ann-Dorthe Zwisler, Rod Taylor, Jørgen Feldbæk Nielsen, Lars Tang. Non-pharmacological rehabilitation interventions for survivors of cardiac arrest: a systematic review and meta-analysis. PROSPERO 2018 CRD42018110129 Available from: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42018110129

Review question

What is the effectiveness of non-pharmacological rehabilitation interventions on adult cardiac arrest survivors?

Searches

The following electronic databases will be searched: MEDLINE, EMBASE, CINAHL, PsycINFO, AMED, The Cochrane Library (Cochrane Database of Systematic Reviews, Cochrane Central Register of Controlled trials (CENTRAL) and Web of Science (up to April 2021 without any restriction in publication date or language).

Trial registries ISRCTN and the WHO ICTRP search portal, and the database ClinicalTrials.gov will be used to search ongoing trials relevant to the review, identify unpublished work, and describe upcoming publications within the studied area. The Cochrane Database of Systematic Reviews will be searched to identify former reviews that identify possible studies for inclusion. These will be screened for eligibility in the same manner as all other studies identified during the database searches.

The search matrix will consist of a combination of relevant indexed terms (e.g. MeSH, Subject Headings or Thesaurus terms), keywords and synonyms for: 1) cardiac arrest, and 2) non-pharmacological rehabilitation interventions.

Title and abstracts from the following conferences will be hand searched: 'European Resuscitation Council Congress', 'American Heart Association' and American College of Cardiology (all from 2009), and the 'Post-cardiac arrest conference' (from 2013).

To identify studies that were not captured by the search matrix, forward and backward citation searching will be conducted on all papers included after full-text screening by two authors (VLJ and MA). Backward citation searching involves screening the references in papers identified after full-text screening. Forward citation searching, searches papers that have cited any papers found after full test screening. References will be hand-searched and citations will be searched for via Web of Science. Titles will be screened for eligibility. Abstracts of possible eligible studies will be screened for eligibility as per all other studies identified during the original database searches.

The first and last authors of unobtainable studies or studies with missing data will be contacted.

Types of study to be included

Included:

- Randomised controlled trials using individual, cluster or any design including parallel group, cross overallocation trials, including pilot studies. - Non-randomised controlled trials

- Prospective and retrospective observational studies with or without a control group.

Excluded:

- Case reports

Condition or domain being studied

The number of people surviving a cardiac arrest is increasing due to improvements in bystander resuscitation and acute hospital care. However, following a cardiac arrest up to half of survivors experience cognitive, psychological and physical problems. These secondary problems plus the underlying cardiac condition present in the majority of cardiac arrest survivors may impact on survivors' well-being and healthrelated quality of life. Non-pharmacological rehabilitation interventions have been recommended in international guidelines but the effect of these interventions remains unknown. A recent investigation into the outcomes for testing effectiveness of interventions for cardiac arrest survivors, by the COSCA initiative (core outcome set for cardiac arrest), recommended three measures be used: survival at 30 days or hospital discharge, neurological function and health-related quality of life (1).

For reference list, see additional information section.

Participants/population

Studies investigating adults, over the age of 18, of both sexes, and all ethnicities who have survived a cardiac arrest will be eligible for inclusion.

Studies that include both survivors of cardiac arrest, and people with cardiac disease without cardiac arrest will be managed in the following way; all trials that present data for cardiac arrest survivors in a subgroup will be included. Where data is not presented in a sub-group, we will contact trial authors to ascertain separate data on the survivors of cardiac arrest. If it is not possible to ascertain subgroup data and only pooled data is available, the study will only be included if at least 50% of participants were survivors of a cardiac arrest.

Intervention(s), exposure(s)

Studies investigating any non-pharmacological rehabilitation intervention for survivors of cardiac arrest, will be considered eligible for inclusion. Non-pharmacological interventions refers to interventions that are not primarily surgically or pharmacologically based or do not involve invasive technology. There will be no restrictions related to length of intervention, timing of intervention or timing of follow-up. The intervention may be a single session or a series of sessions. It may be provided one-to-one or in a group of survivors. Most cardiac arrest survivors have an underlying cardiac condition and hence cardiac rehabilitation intervention studies that include cardiac arrest survivors will be considered as a non-pharmacological intervention and included in this review. The cardiac rehabilitation may be exercise-based, psychological-based, education based or comprehensive in nature.

Exclusion criteria:

1. All studies reporting on effectiveness of medical, surgical or invasive technology interventions will be excluded.

2. Interventions in the emergency room or critical care unit will be excluded.

3. Studies that do not report results at baseline and at minimum one follow-up point after intervention.

Comparator(s)/control

Comparator can include no treatment, active control, usual care, or where the intervention is in addition to another non-pharmacological intervention (for example, as an add on to exercise rehabilitation) or no comparator.

Context

The intervention may take place in any setting: in the hospital (but not while in the emergency room/critical care unit), outside the hospital, at survivors' home or may start in hospital and continue following discharge from hospital or via telemedicine.

Main outcome(s)

The outcomes are based on the those recommended by the COSCA initiative (1). The primary outcome will be health-related quality of life. Measures of health-related quality of life outcomes will include generic or disease-specific patient reported outcome measures. The outcome measure can be a single-item tool (e.g. 'How would you rate your overall quality of life?') or a multi item tool (examples include 36-item Short Form Health survey and EuroQoL five dimensions questionnaire). For multi component/dimensional outcome measures a subscale which contains health-related quality of life will be favoured over the overall score of the measurement even if the overall score reflects health-related quality of life.

Main secondary outcome: neurological function. Measures of neurological function measure the level of disability after a neurological event. Included measures may primarily test cognitive ability (for example the Mini-mental state examination) or may combine cognitive and physical ability hence measuring global disability (for example the Modified Rankin Scale). Measures may be clinician reported (for example the Cerebral Performance Category or Glasgow Outcome Scale –Extended) or observer reported (for example the Informant Questionnaire on Cognitive Decline in the Elderly, completed by relatives or carers) or patient reported (for example Two-simple questions).

Measures of effect

It has been suggested that outcomes for cardiac arrest survivors evolve over time and survivors should be reassessed at 30-days, 60-days and one-year after arrest (1, 2). Hence, data will be extracted at all time points that are the nearest possible to the recommended follow-up points (1) (survival at 30 days or hospital discharge, three-month follow-up and longest follow up point) to discern if outcomes change over time.

Additional outcome(s)

Secondary outcomes:

1. Survival at 30 days or hospital discharge (if both are reported, survival at 30 days will be favoured over survival at hospital discharge), and one year. Mortality due to any cause, but if proportion due to cardiac cause is available this will be reported.

2. Re-hospitalization (all cause and the proportion due to cardiac cause if available)

3. Serious adverse events: defined as resulting in death or re-hospitalization causing significant disability ,including cardiovascular complications such as cardiac arrest or any arrhythmia with hemodynamic compromise.

4. Non-serious adverse events for example musculoskeletal injury, palpitations, or dizziness.

5. Psychological well-being: measured by patient reported symptoms of anxiety, depression, post-traumatic stress disorder or stress. The outcome measure can be a single-item tool (for example 'How would you rate your overall psychological well-being?') or a multi item tool (for example the Hospital Anxiety and Depression Scale).

6. Fatigue outcomes: measured by patient reported outcomes such as the Fatigue impact scale. Measures may be disease specific or generic and multi-component/dimensional. For multi component/dimensional outcome measures a sub-scale which contains fatigue will be favoured over the overall score of the measurement even if the overall score reflects fatigue.

7. Exercise capacity: measured by aerobic fitness. This will include any objective measure of the ability of the heart and lungs to get oxygen to the muscles where it can be consumed. This could be maximal or peak oxygen consumption. A change of the aerobic fitness could be an increase in peak oxygen consumption (VO2peak) obtained from a maximal cardiopulmonary exercise test or as a decrease in sub-maximal oxygen uptake at a given work load, or a decrease of sub-maximal heart rate at a given work load.

8. Physical capacity: measured by self-reported questionnaires, single item questions or objectives measures for example activity monitors or step counters.

Data extraction (selection and coding)

Selection of studies will be done by merging all search results into the technology platform, Covidence. Duplicates will be removed before two authors (VLJ and MA) will independently screen titles and abstracts followed by full-text screening of potentially eligible studies. Any disagreements will be discussed and resolved with a third review author (JC). For randomised controlled crossover trials data will be handled as it would have been a randomised controlled trial and therefore data will be extracted for baseline and from the assessment of effect from the first period (data from after the cross-over will not be extracted).

A standardised pre-piloted form will be used to extract data from all the included studies.

The following data will be extracted if it is relevant in terms of study design (observational vs randomised controlled studies) by two independent reviewers (VLJ & MA):

1. Source of study and author contact details

2. Study design, study duration, setting and country.

3. Participant characteristics: age, sex, cause of cardiac arrest, place of cardiac arrest, cardiac arrest circumstances, health-status of participants including details of any ongoing cardiac condition (for example myocardial infarction), in hospital interventions and whether they received targeted temperature management and/or an implantable cardioverter defibrillator.

4. Number of groups and number of participants in each study, and study arm.

5. Description and components of rehabilitation intervention and any control, length of intervention, dose and frequency.

6. Theory or mechanism of the study intervention.

7. Which primary and secondary outcomes as defined above are present and time points of outcomes.

8. For each outcome of interest: sample size, estimate of effect and confidence interval; p value and subgroup analyses.

9. Information for assessment of risk of bias.

Any discrepancies in data extraction will be investigated and discussed, then if necessary resolved with a third review author (JC).

Risk of bias (quality) assessment

Two review authors (VLJ & MA) will independently assess the risk of bias in included studies. RoB 2.0 tool for randomised controlled trials ROBINS-I (risk of bias in non-randomised Studies – of Interventions) will be used to assess risk of bias in observational studies.

Risk of bias will be assessed for each outcome within each study. While some items are generic across outcomes, where potential differences may occur (for example between subjective and objective measures), outcomes will be assessed separately and may be judged at a different level of bias within the same study.

Any disagreements between review authors will be discussed and resolved with a third review author (JC).

Strategy for data synthesis

Results from the different study designs will be presented and pooled separately.

We will undertake a meta-analysis where two or more trials are similar enough clinically and statistically for pooling of trials to be appropriate. Where there is high heterogeneity between studies or inappropriate quantitative reporting of outcomes, a narrative synthesis of outcomes from included studies will be provided.

If it is possible to conduct a meta-analysis a random effects meta-analysis will be used given the likely presence of some clinical heterogeneity across studies. Continuous data will be expressed as the mean difference (MD) or standardized mean difference (SMD) and their respective confidence intervals (CI) will be calculated.

For dichotomous outcomes (serious and non-serious adverse events) and for outcomes of observational studies relative risk ratios (RR) along with a CI will be calculated using random effects meta-analysis.

For time to even outcome (survival, re-hospitalization) hazard ratios will be pooled if presented.

Statistical heterogeneity of the study results will be examined using Cochran Q test and quantified with I^2 values and the between study variance I^2. Qualitative assessment of heterogeneity will be assessed by comparing the characteristics of included studies. Assessment of small study bias will be assessed by calculating an Egger's test score and illustrated with a funnel plot. If small study bias is present defined by a positive Egger's test, a metatrim analysis will be conducted.

If the included studies need network meta-analysis to be pooled this will be performed, as described in chapter 16.6.3 of the Cochrane Handbook.

The confidence in estimates of the effects of interventions will be assessed using the Grading of Recommendation, Assessment, Development and Evaluation system (GRADE).

The paper will be reported according to the Preferred Reporting Items for Systematic Reviews and MetaAnalyses (PRISMA) guidelines.

Analysis of subgroups or subsets

If possible, we will carry out the following subgroup and stratified analyses to explore heterogeneity of the studies. Location of cardiac arrest (in versus out-of-hospital).

1. Mode of delivery of intervention (supervised vs non-supervised).
2. Provision of intervention (individual vs group based).
3. Content of rehabilitation, single component (for example: exercise-based/education-based/psychological based) vs comprehensive.
4. Received an implantable cardioverter defibrillator vs did not receive an implantable cardioverter defibrillator.
5. Random sequence generation (low/some concerns/high); random sequence concealment (low/some concerns/high).
6. Overall risk of bias (low/some concerns/high).
7. Length of intervention (single session vs 1-6-weeks vs over 6-weeks) 8. Duration of trial follow-up (1-12 weeks vs 13-24 weeks vs over 24 weeks).
9. Setting of trials (single vs multicentre).
10. Continent of publication.
11. Self-reported cognitive ability vs self-reported cognitive and physical ability.

12. Clinician reported cognitive ability vs observer reported vs self-reported.

Meta-regression will be carried out to investigate the effect of continuous variables including age, sex distribution and all sub-group analyses listed above.

Contact details for further information

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Organisational affiliation of the review

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Type and method of review

Intervention, Meta-analysis, Systematic review

Anticipated or actual start date

01 November 2018

Anticipated completion date

31 July 2021

Funding sources/sponsors

Funding for the writing of the review protocol, data analysis and preparation of the article for publication will be provided by REHPA - Danish Knowledge Centre for Rehabilitation and Palliative Care, University of Southern Denmark.

Conflicts of interest

Language

English

Country

Denmark

Stage of review

Review Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Heart Arrest; Humans; Medicine; Survivors

Date of registration in PROSPERO
11 October 2018

Date of first submission
19 September 2018

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	Yes
Formal screening of search results against eligibility criteria	Yes	Yes
Data extraction	Yes	Yes
Risk of bias (quality) assessment	Yes	Yes
Data analysis	Yes	Yes

The record owner confirms that the information they have supplied for this submission is accurate and complete and they understand that deliberate provision of inaccurate information or omission of data may be construed as scientific misconduct.
The record owner confirms that they will update the status of the review when it is completed and will add publication details in due course.

Versions
11 October 2018
07 May 2021
12 May 2021
15 May 2021

Supplementary data Table 1. Search Strategy

MEDLINE	
POPULATION	INTERVENTION
heart arrest [MeSH Terms] out-of-hospital cardiac arrest [MeSH Terms] heart arrest [Title/Abstract] heart arrests [Title/Abstract] cardiac arrest [Title/Abstract] cardiac arrests [Title/Abstract] cardiopulmonary arrest [Title/Abstract]	rehabilitation [MeSH Terms] physical medicine [MeSH Terms] rehabilitation nursing [MeSH Terms] exercise [MeSH Terms] social support [MeSH Terms] psychological adaptation [MeSH Terms] cognitive behavior therapy [MeSH Terms] health education [MeSH Terms] aftercare [MeSH Terms] rehabilitation, vocational [MeSH Terms] rehabilitation [Title/Abstract] vocational [Title/Abstract] aftercare [Title/Abstract] telerehabilitation [Title/Abstract] physical medicine [Title/Abstract] exercise [Title/Abstract] exercises [Title/Abstract] physical activity [Title/Abstract] social support [Title/Abstract] psychological adaptation [Title/Abstract]) coping behavior [Title/Abstract]) coping skills [Title/Abstract]) adaptive behavior [Title/Abstract]) cognitive behavior therapy [Title/Abstract] cognitive behavioral therapy [Title/Abstract] cognitive behavior therapies [Title/Abstract] cognitive behavioral therapies [Title/Abstract] cognitive psychotherapy [Title/Abstract]
	cognitive psychotherapies [Title/Abstract] acceptance and commitment therapy [Title/Abstract] mindfulness [Title/Abstract] health education [Title/Abstract])
AMED 1985 to date	
POPULATION	INTERVENTION
(MH “Heart arrest” explode)	(MH "Rehabilitation+") (MH "Physical Medicine") (MH “Rehabilitation nursing”) (MH Exercise+) (MH Support, Psychosocial+)

	(MH Adaptation, Psychological+) (MH Cognitive therapy+) (MH Health education+) (MH After Care)
TI OR AB	TI OR AB
Heart arrests Cardiac arrest Cardiac arrests Cardiopulmonary arrest	Rehabilitation Vocational Aftercare Telerehabiliation Physical medicine Exercise Exercises Physical activity Social support psychosocial psychological adaptation coping behavior coping skills adaptive behavio* cognitive behavio#r therap* cognitive behavio#ral therap* cognitive behavioral therap* cognitive psychotherap* acceptance and commitment therapy mindfulness health education
CINAHL	
POPULATION	INTERVENTION
(MH "Heart arrest" explode)	(MH "Rehabilitation+") (MH "Physical Medicine") (MH "Rehabilitation nursing") (MH Exercise+) (MH Support, Psychosocial+) (MH Adaptation, Psychological+) (MH Cognitive therapy+) (MH Health education+) (MH After Care)
TI OR AB	TI OR AB
Heart arrests Cardiac arrest Cardiac arrests Cardiopulmonary arrest	Rehabilitation Vocational Aftercare Telerehabiliation Physical medicine Exercise

	Exercises Physical activity Social support psychosocial psychological adaptation coping behavior coping skills adaptive behavio* cognitive behavio#r therap* cognitive behavio#ral therap* cognitive behavioral therap* cognitive psychotherap* acceptance and commitment therapy mindfulness health education
Embase 1974 to present	
heart arrest Exp out-of-hospital cardiac arrest Exp	Rehabilitation (Exp all) Physical medicine Rehabilitation nursing Exercise Social support Coping behavior Cognitive behavioral therapy Health education
Ti OR Ab Heart arrests Cardiac arrest Cardiac arrests Cardiopulmonary arrest	Ti or Ab Rehabilitation Vocational Aftercare Telerehabiliation Physical medicine Exercise Exercises Physical activity Social support psychosocial psychological adaptation coping behavior coping skills adaptive behavio* cognitive behavio#r therap* cognitive behavio#ral therap* cognitive behavioral therap* cognitive psychotherap* acceptance and commitment therapy mindfulness health education

PsychINFO 1806 to present	
All fields: Heart arrest Heart arrests Cardiac arrest Cardiac arrests Cardiopulmonary arrest	Subject headings: Rehabilitation Exercise Social support Cognitive therapy Health education Ti or Ab Rehabilitation Vocational Aftercare Telerehabiliation Physical medicine Exercise Exercises Physical activity Social support psychosocial psychological adaptation coping behavior coping skills adaptive behavio* cognitive behavio#r therap* cognitive behavio#ral therap* cognitive behavioral therap* cognitive psychotherap* acceptance and commitment therapy mindfulness health education
Web of Science	
Heart arrest Heart arrests Cardiac arrest Cardiac arrests Cardiopulmonary arrest <u>Search string</u>: (TS=Topic) TS=(rehabilitation) OR TS=(vocational) OR TS=(aftercare) OR TS=(telerehabilitation) OR TS=("physical medicine") OR TS=(exercise) OR TS=(exercises) OR TS=("physical activity") OR TS=("social support") OR TS=("psychological adaptation") OR TS=("coping behavio*") OR TS=("coping skills") OR TS=("adaptive behavio*") OR TS=("cognitive behavio*	Rehabilitation Vocational Aftercare Telerehabiliation Physical medicine Exercise Exercises Physical activity Social support psychosocial psychological adaptation coping behavior coping skills adaptive behavio* cognitive behavio#r therap*

therap*") OR TS=("cognitive psychotherap*") OR TS=("acceptance and commitment therapy") OR TS=(mindfulness) OR TS=("health education") AND TS=("heart arrest") OR TS=("cardiac arrest") OR TS=("heart arrests") OR TS=("cardiac arrests") OR TS=("cardiopulmonary arrest") OR TS=("cardiopulmonary arrests")	cognitive behavio#ral therap* cognitive behavioral therap* cognitive psychotherap* acceptance and commitment therapy mindfulness health education
The Cochrane library (SRs and CENTRAL)	
As for Medline but using CENTRAL's search builder syntax.	
International Standard Randomised Controlled Trial Number (ISRCTN)	
Cardiac arrest	
World Health organisation International Clinical Trials Registry Platform (ICTRP) and ClinicalTrials.gov	
(Condition) Cardiac arrest	OR (Other terms) Rehabilitation Vocational Aftercare Telerehabiliation Physical medicine Exercise Physical activity Social support psychosocial psychological adaptation coping behavior cognitive behavio#r therap* cognitive behavio#ral therap* cognitive behavioral therap* cognitive psychotherap* acceptance and commitment therapy mindfulness health education

Sample search matrix:

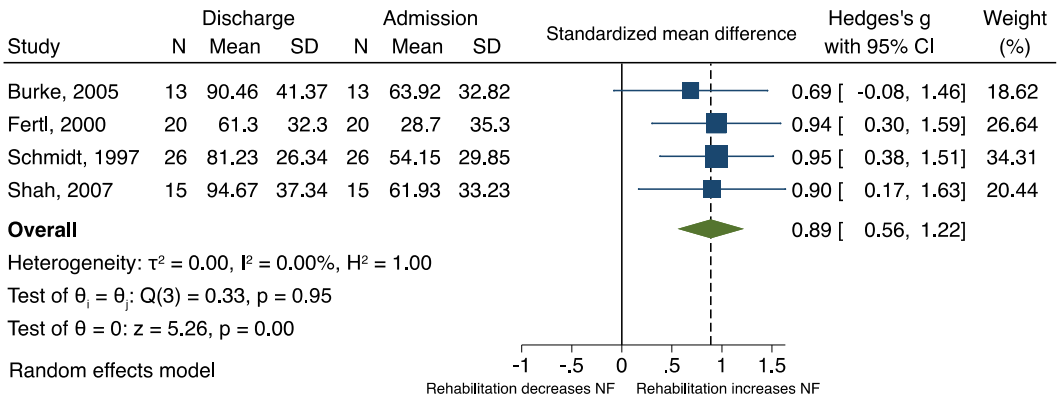
MEDLINE Search matrix:

Search ((rehabilitation[MeSH Terms]) OR physical medicine[MeSH Terms]) OR rehabilitation nursing[MeSH Terms]) OR exercise[MeSH Terms]) OR social support[MeSH Terms]) OR psychological adaptation[MeSH Terms]) OR cognitive behavior therapy[MeSH Terms]) OR health education[MeSH Terms]) OR aftercare[MeSH Terms]) OR rehabilitation, vocational[MeSH Terms]) OR rehabilitation[Title/Abstract]) OR vocational[Title/Abstract]) OR aftercare[Title/Abstract]) OR telerehabilitation[Title/Abstract]) OR physical medicine[Title/Abstract]) OR exercise[Title/Abstract]) OR exercises[Title/Abstract]) OR physical activity[Title/Abstract]) OR social support[Title/Abstract]) OR psychological adaptation[Title/Abstract]) OR coping behaviour[Title/Abstract]) OR coping skills[Title/Abstract]) OR adaptive behavior[Title/Abstract]) OR cognitive behavior therapy[Title/Abstract]) OR cognitive behavioral therapy[Title/Abstract]) OR cognitive behavior therapies[Title/Abstract]) OR cognitive behavioral therapies[Title/Abstract]) OR cognitive psychotherapy[Title/Abstract]) OR cognitive psychotherapies[Title/Abstract]) OR ((acceptance[Title/Abstract] AND commitment therapy[Title/Abstract]))) OR mindfulness[Title/Abstract]) OR health education[Title/Abstract])) AND (((((((heart arrest[MeSH Terms]) OR out of hospital cardiac arrest[MeSH Terms]) OR heart arrest[Title/Abstract]) OR heart arrests[Title/Abstract]) OR cardiac arrest[Title/Abstract]) OR cardiac arrests[Title/Abstract]) OR cardiopulmonary arrest[Title/Abstract])

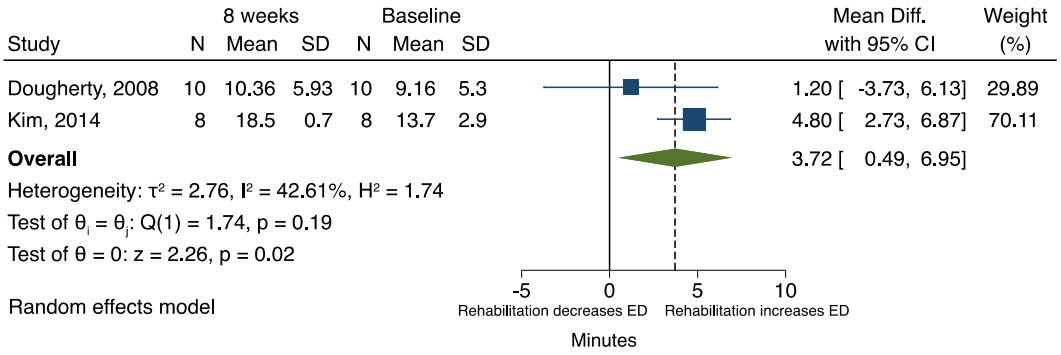
Supplementary data Table 2. Studies excluded at full text stage with reasons

Excluded study	**Reason for exclusion**
Ada, 2018	Conference abstract
Arabia, 2011	Mixed population of people who have suffered a major cardiac event
Baston, 2017	Conference abstract
Berg, 2020	Mixed population of ICDs implanted for primary and secondary prevention and authors did not have separate data on number of cardiac arrest survivors.
Bermejo, 2015	Conference abstract
Boyce, 2017	Outcome data only at baseline
Chanu, 2016	Not survivor of cardiac arrest population
Helmark, 2016	Conference abstract
Choi, 2017	No survivor of cardiac arrest population
Dougherty, 1997	Not an intervention study
Dougherty, 2015	Mixed population of ICDs implanted for primary and secondary prevention and authors did not have separate data on number of cardiac arrest survivors.
Exposito, 2012	Conference abstract
Goldman, 2013	Conference abstract
Harbinson, 2017	Not survivor of cardiac arrest population
Irvine, 2011	Not survivor of cardiac arrest population
Kim, 2017	Not effect study
Ko, 2020	Not survivor of cardiac arrest population
Konh, 2000	Not survivor of cardiac arrest population
Markus, 2017	Not rehabilitation intervention
Mochizuki 2014	Conference abstract
Moroni, 2006	Not survivor of cardiac arrest population
Moulaert, 2016	Economic evaluation
Moulaert, 2011	Study rationale
Moulaert, 2007	Study protocol
Munjal, 2018	Conference abstract
Sears, 2004	Systematic review
Stock, 2019	Not survivor of cardiac arrest population
Takahashi, 2015	Intervention in intensive care unit

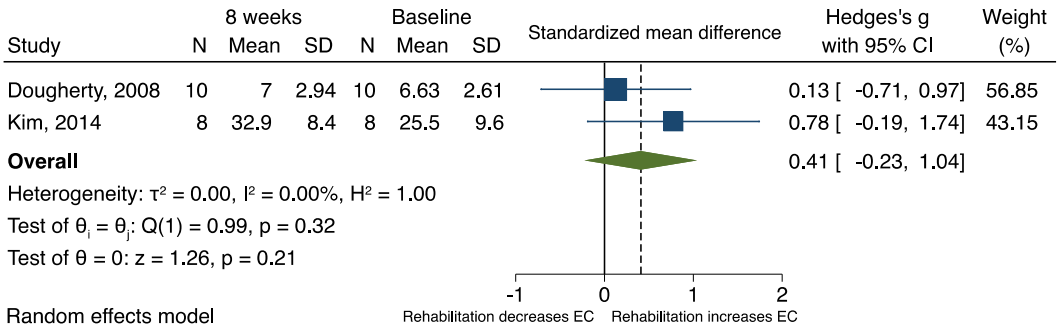
Supplementary data Fig. 1. Forest plot for effect of inpatient rehabilitation for CA-survivors with acquired brain injury on neurological function (NF) between admission and discharge, sensitivity analysis with Howell (2013) removed due to heterogeneity in study population and presence of statistical heterogeneity.



Supplementary data Fig. 2. Forest plot for effect of exercise-based rehabilitation for CA-survivors on exercise duration (ED) (minutes) between baseline and 8 weeks follow-up.



Supplementary data Fig. 3. Forest plot for effect of exercise-based rehabilitation for CA-survivors on exercise capacity (EC) between baseline and 8 weeks follow-up.



Supplementary data Table 3. Summary of findings: HRQoL

Rehabilitation for improving health related quality of life (HRQoL) in survivors of cardiac arrest					
Randomised controlled trials					
Outcome	Standard care	Rehabilitation	Number of participants (studies)	Quality of evidence[a] (GRADE)	Comments
HRQoL Physical component score difference between baseline and 12 months follow-up (0-100 points, higher scores better)	12.8	mean **4.75** points greater compared to standard care	108(2)	++oo Low[b]	
HRQoL Mental component score difference between baseline and 12 months follow-up (0-100, higher scores better)	7.57	mean **3.26** points greater compared to standard care	108(2)	++oo Low[b]	
Prospective observational studies					
HRQoL Physical component score difference between at 6 months follow-up (0-100, higher scores better)	-	mean **20.32** point increase	82(2)	+ooo Very low[c,d]	No comparison arm included in either trial
HRQoL Mental component score at 6 months follow-up (0-100, higher scores better)	-	mean **16.76** point increase	82(2)	+ooo Very low[c]	No comparison arm included in either trial
[a]GRADE Working Group grades of evidence **High quality**: Further research is very unlikely to change our confidence in the estimate of effect. **Moderate quality**: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. **Low quality**: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. **Very low quality**: We are very uncertain about the estimate.					
Explanations [b]High risk of bias in one of the two studies ('some concerns' with criteria: deviations from intended outcomes and selection of the reported results, 'high risk of bias' with criteria: measurement of the outcome), some indirectness of evidence in one study (intervention aimed at people with new ICD implanted) and due to the serious imprecision in both studies (small number of participants). [c]High risk of bias in one study out of two studies, both studies were observational. [d]Considerable heterogeneity (I^2=90.17%) (point estimates and confidence intervals vary considerably).					

Key: CI: Confidence interval; HRQoL: Health related quality of life; SMD: Standardized mean difference

Supplementary data Table 4. Summary of findings: neurological function

Effect of inpatient rehabilitation on neurological function for survivors of cardiac arrest with acquired brain injury					
All observational studies					
Outcome follow-up	Standard care	Rehabilitation	Number of participants (studies)	Certainty of evidence (GRADE[a])	Comments
Improvement in function between admission and discharge (Functional independence measure and Barthel index)	-	SMD **0.71** effect size (CI 0.45-0.96)	187(5)	+ooo Very Low[d]	No comparison arm included in any included trial
[a]GRADE Working Group grades of evidence **High quality**: Further research is very unlikely to change our confidence in the estimate of effect. **Moderate quality**: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. **Low quality**: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. **Very low quality**: We are very uncertain about the estimate					
[d]High risk of bias in all studies (multiple domains) and all were observational studies.					

Key: CI: Confidence interval; SMD: Standardized mean difference